

31/5-7

Composite Log

Company	Equinor
Well	31/5-7
Field	Eos
Nation	Norway
Continent	Europe
Location	Norwegian North Sea
Field Loc. 1 / Northing	Well Composite: 8 1/2 in. pilot, 26 in, 12 1/4 in pilot, 12 1/4 in and 8.5 in
Latitude	060 34' 35.140" N DMS
Longitude	003 26' 36.130" E DMS
Permanent Datum	MSL
Elevation	0.00 M
Elevations K.B.	0.00 M
Elevations D.F.	31.00 M
Elevations G.L.	-307.00 M
Elev. Log Zero	31.00 M
Above Permanent Datum	31.00 M
Log Measured From	DF
Drill Measured From	DF
Service Order	19SCA0085
Spud Date	02-Dec-2019
Depth	2915
Date Plotted	Wednesday, 5 February 2020
Time Plotted	9:34:47 AM

PETROLOG
ADVANCED LOG ANALYSIS SOFTWARE

PETROLOG SOFTWARE VERSION 10.7.1.5

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ADVANCED LOG ANALYSIS SOFTWARE

Since well log interpretations are opinions based upon inferences from well logs, we cannot and do not guarantee the correctness or accuracy of any interpretation. Therefore we shall not be liable or responsible for any loss, damage, cost or expense incurred or sustained by anyone resulting from any interpretation.

Run Information					
Run Number	Run 2	Run 3	Run 5	Run 8	Run 9
Date	04-Dec-2019	05-Dec-2019	16-Dec-2019	23-Dec-2019	29-Dec-2019
Depth Driller	944.00 M	931.00 M	1906.00 M	2584.00 M	2592.00 M
Depth Logger	944.00 M	931.00 M	1906.00 M	2584.00 M	2592.00 M
First Reading	936.20 M	906.80 M	1896.80 M	2574.80 M	2591.22 M
Last Reading	406.80 M	406.60 M	925.80 M	1900.20 M	2584.70 M
Casing Driller	406.60 M	406.60 M	925.10 M	1896.70 M	2583.00 M
Casing Logger	406.60 M	406.60 M	925.10 M	1896.70 M	2583.00 M
Casing Size	30.000 INCH	30.000 INCH	20.000 INCH	13.625 INCH	9.625 INCH
Casing Weight	457.00 LB/F3	457.00 LB/F3	133.00 LB/F3	59.27 LB/F3	53.50 LB/F3
Bit Size	8.500 INCH	26.000 INCH	12.250 INCH	12.250 INCH	8.500 INCH
Hole Fluid Type	Seawater	Seawater	Glydril	Glydril	Glydril
Density	1.030 G/CC	1.030 G/CC	1.250 G/CC	1.260 G/CC	1.150 G/CC
pH	-1E+30	-1E+30	-1E+30	-1E+30	-1E+30
Sample Source	Active Tank	Active Tank	Active Tank	Active Tank	Active Tank
Rm @ Measured Temperature	0.190 OHMM	0.190 OHMM	0.076 OHMM	0.070 OHMM	0.070 OHMM
Mud temp @ Surface	23.90 DEGC	23.90 DEGC	17.78 DEGC	17.10 DEGC	18.00 DEGC
Rmf @ Measured Temperature	0.190 OHMM	0.190 OHMM	0.070 OHMM	0.060 OHMM	0.060 OHMM
MF temp @ Surface	23.90 DEGC	23.90 DEGC	18.60 DEGC	18.06 DEGC	18.00 DEGC
RMC @ Measured Temperature			0.090 OHMM	0.100 OHMM	0.100 OHMM
MC temp @ Surface	20.00 DEGC	20.00 DEGC	20.00 DEGC	20.00 DEGC	20.00 DEGC
Source Rmc	Pressed	Pressed	Pressed	Pressed	Pressed
Rm @ BHT	0.230 OHMM	0.230 OHMM	0.054 OHMM	0.041 OHMM	0.046 OHMM
Rm temp @ Bottom	23.90 DEGC	23.90 DEGC	17.78 DEGC	17.10 DEGC	18.00 DEGC
Rmf @ BHT	0.230 OHMM	0.230 OHMM	0.051 OHMM	0.036 OHMM	0.040 OHMM

Rmf temp @ Bottom	23.90 DEGC	23.90 DEGC	18.60 DEGC	18.06 DEGC	18.00 DEGC
Rmc @ BHT			0.067 OHMM	0.062 OHMM	0.070 OHMM
Rmc temp @ Bottom	20.00 DEGC	20.00 DEGC	20.00 DEGC	20.00 DEGC	20.00 DEGC
Time Since Circulation	20:23:38	02:10:48	14:10:12	13:49:45	05:04:18
Date circ. stopped	04-Dec-2019	07-Dec-2019	17-Dec-2019	25-Dec-2019	30-Dec-2019
Tool on Bottom	00:53:56	01:03:12	08:19:22	06:55:46	06:12:03
Date logger at btm	04-Dec-2019	05-Dec-2019	16-Dec-2019	23-Dec-2019	29-Dec-2019
Surface hole temp	10.00 DEGC	20.00 DEGC	4.00 DEGC	4.00 DEGC	6.00 DEGC
Bottom Hole Temperature	16.00 DEGC	16.00 DEGC	34.00 DEGC	45.00 DEGC	38.00 DEGC
Surface temperature	10.00 DEGC	20.00 DEGC	4.00 DEGC	4.00 DEGC	6.00 DEGC
Maximum Recorded Temperature	16.00 DEGC	16.00 DEGC	34.00 DEGC	45.00 DEGC	38.00 DEGC
Max recorded temp 1	16.00 DEGC	16.00 DEGC	34.00 DEGC	45.00 DEGC	38.00 DEGC
Equipment No.	A615	A615	A615	A615	A615
Base Location	Service Office	Service Office	Service Office	Service Office	Service Office
Logging Company ID	440	440			
Recorded By	J.Johansen	J.Johansen	G.Boyd	G.Boyd	S.Kroevel
Witnessed By	R.Nordstokke	S.Vikra	S.Vikra	J.Eikenes	J.Eikenes
Program Version	20C0-999	20C0-999	20C0-999	20C0-999	20C0-999
Fluid Level	31.00 M	31.00 M	3.00 M	3.00 M	3.00 M
General.Temp Select	GR_SURF	GR_SURF	GR_SURF	GR_SURF	GR_SURF
Maximum Hole Deviat	0.18 DEG	1.12 DEG	1.02 DEG	0.26 DEG	1.10 DEG
Depth	2915.00	2915.00	2915.00	2915.00	2915.00
Temp Conn.Water Samp.	20.00 DEGC	20.00 DEGC	20.00 DEGC	20.00 DEGC	20.00 DEGC
Rig Type	Semi-Submersible	Semi-Submersible	Semi-Submersible	Semi-Submersible	Semi-Submersible
Conveyance Type	Drill Pipe (TLC)	Drill Pipe (TLC)	Drill Pipe (TLC)	Drill Pipe (TLC)	Drill Pipe (TLC)
Geothermal Gradient	12	18.22689	18.22689	15	14
Generalized Mud Resistivity Sel	GEN9	GEN9	GEN9	GEN9	GEN9
Generalized Caliper Selection	BS	BS	BS	BS	BS
Form Factor Exponent	2	2	2	2	2
Form Factor Numerator	1	1	1	1	1
Form Factor Porosity Source	PHI	PHI	PHI	PHI	PHI
Matrix Density	2.71	2.71	2.71	2.71	2.71
Fluid Density	1	1	1	1	1
Integrated Hole Volume Caliper	BS	BS	BS	BS	BS
Integrated Hole Volume Control	START	START	START	START	START
Resistivity of Connate Water	1	1	1	1	1
Borehole Salinity	33579.09	33579.09	114390.2	130260.8	93984
Run Number	Run 10	Run 12	Run 13		
Date	31-Dec-2019	07-Jan-2020	13-Jan-2020		
Depth Driller	2643.00 M	2710.00 M	2915.00 M		
Depth Logger	2643.00 M	2710.00 M	2915.00 M		
First Reading	2642.28 M	2709.03 M	2914.28 M		
Last Reading	2584.90 M	2630.90 M	2699.00 M		
Casing Driller	2583.00 M	2583.00 M	2583.00 M		
Casing Logger	2583.00 M	2583.00 M	2583.00 M		
Casing Size	9.625 INCH	9.625 INCH	9.625 INCH		
Casing Weight	53.50 LB/F3	53.50 LB/F3	53.50 LB/F3		
Bit Size	8.500 INCH	8.500 INCH	8.500 INCH		
Hole Fluid Type	Glydril	Glydril	Glydril		
Density	1.150 G/CC	1.150 G/CC	1.150 G/CC		
pH	-1E+30	-1E+30	-1E+30		
Sample Source	Active Tank	Active Tank	Active Tank		
Rm @ Measured Temperature	0.070 OHMM	0.080 OHMM	0.090 OHMM		
Mud temp @ Surface	15.00 DEGC	14.80 DEGC	18.90 DEGC		
Rmf @ Measured Temperature	0.060 OHMM	0.070 OHMM	0.070 OHMM		
MF temp @ Surface	20.00 DEGC	14.80 DEGC	17.90 DEGC		
RMC @ Measured Temperature	0.090 OHMM	0.090 OHMM	0.100 OHMM		
MC temp @ Surface	15.00 DEGC	20.00 DEGC	20.00 DEGC		
Source Rmc	Pressed	Pressed	Pressed		
Rm @ BHT	0.040 OHMM	0.044 OHMM	0.050 OHMM		
Rm temp @ Bottom	15.00 DEGC	14.80 DEGC	18.90 DEGC		
Rmf @ BHT	0.039 OHMM	0.039 OHMM	0.038 OHMM		
Rmf temp @ Bottom	20.00 DEGC	14.80 DEGC	17.90 DEGC		
Rmc @ BHT	0.051 OHMM	0.057 OHMM	0.057 OHMM		
Rmc temp @ Bottom	15.00 DEGC	20.00 DEGC	20.00 DEGC		
Time Since Circulation	07:38:34	17:16:13	10:55:25		
Date circ. stopped	01-Jan-2020	07-Jan-2020	15-Jan-2020		
Tool on Bottom	17:32:56	04:05:33	19:44:45		
Date logger at btm	31-Dec-2019	07-Jan-2020	13-Jan-2020		
Surface hole temp	6.00 DEGC	6.00 DEGC	6.00 DEGC		
Bottom Hole Temperature	43.00 DEGC	44.00 DEGC	51.00 DEGC		
Surface temperature	6.00 DEGC	6.00 DEGC	6.00 DEGC		
Maximum Recorded Temperature	43.00 DEGC	44.00 DEGC	51.00 DEGC		
Max recorded temp 1	43.00 DEGC	44.00 DEGC	51.00 DEGC		
Equipment No.	A615	A615	A615		
Base Location	Service Office	Service Office	Service Office		
Logging Company ID					
Recorded By	S.Kroevel	M.Martter	M.Bergenkull		
Witnessed By	J.Eikenes	R.Nordstokke	R.Nordstokke		
Program Version	20C0-999	20C0-999	20C0-999		

Program Version	2000-999	2000-999	2000-999		
Fluid Level	3.00 M	3.00 M	3.00 M		
General.Temp Select	GR_SURF	GR_SURF	GR_SURF		
Maximum Hole Deviat	1.12 DEG	1.12 DEG	1.12 DEG		
Depth	2915.00	2915.00	2915.00		
Temp Conn.Water Samp.	20.00 DEGC	20.00 DEGC	20.00 DEGC		
Rig Type	Semi-Submersible	Semi-Submersible	Semi-Submersible		
Conveyance Type	Drill Pipe (TLC)	Drill Pipe (TLC)	Drill Pipe (TLC)		
Geothermal Gradient	17	17	17		
Generalized Mud Resistivity Sel	GEN9	GEN9	GEN9		
Generalized Caliper Selection	BS	BS	BS		
Form Factor Exponent	2	2	2		
Form Factor Numerator	1	1	1		
Form Factor Porosity Source	PHI	PHI	PHI		
Matrix Density	2.71	2.71	2.71		
Fluid Density	1	1	1		
Integrated Hole Volume Caliper	BS	BS	BS		
Integrated Hole Volume Control	START	START	START		
Resistivity of Connate Water	1	1	1		
Borehole Salinity	93984	97140	79402		

Logging Remarks:
All depths referenced to driller's depth. See EOWR for depth tracking details.
All data from tool memory. All data acquired while drilling except where noted.
Run 2:
Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and neutron activation.
Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
Bulk Density is compensated for tool standoff/mud cake.
Neutron Porosity is calculated with limestone matrix and is environmentally corrected for bit size, mud weight, temperature, pressure and mud salinity.
Small intervals of missing Neutron Porosity data due to the practice of quickly resuming drilling after connections, before the tool has sufficient time to start PH.
Interval of missing data at 532.6 m bit depth due to low flow when resuming drilling after connection.
Run 2 and 8 1/2 in. pilot section TD at 944 m.
Run 3:
Gamma Ray measurement is corrected for collar thickness only.
Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
Phase Resistivity 16 and 22 in. spacing adversely affected by large borehole corrections.
Run 3 and 26 in. section TD at 931 m.
Run 5:
Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.
Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
Run 5 and 12 1/4 in. pilot section TD at 1906 m.
Run 8:
Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.
Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
Run 8 and 12 1/4 in. section TD at 2584 m.
Run 9-13:
Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.
Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
Run 9 POOH at 2592 m to perform coring.
Run 10: Data acquired while reaming down over cored interval from 2592 to 2596 m bit depth.
Run 10 POOH at 2643 m to perform coring.
Run 12: Data acquired while reaming down over cored interval from 2643 to 2697 m bit depth.
Run 12 POOH at 2710 m to perform coring.
Run 13: Data acquired while reaming down over cored interval from 2710 to 2784 m bit depth.
Run 13, 8 1/2 in. section and well TD at 2915 m.

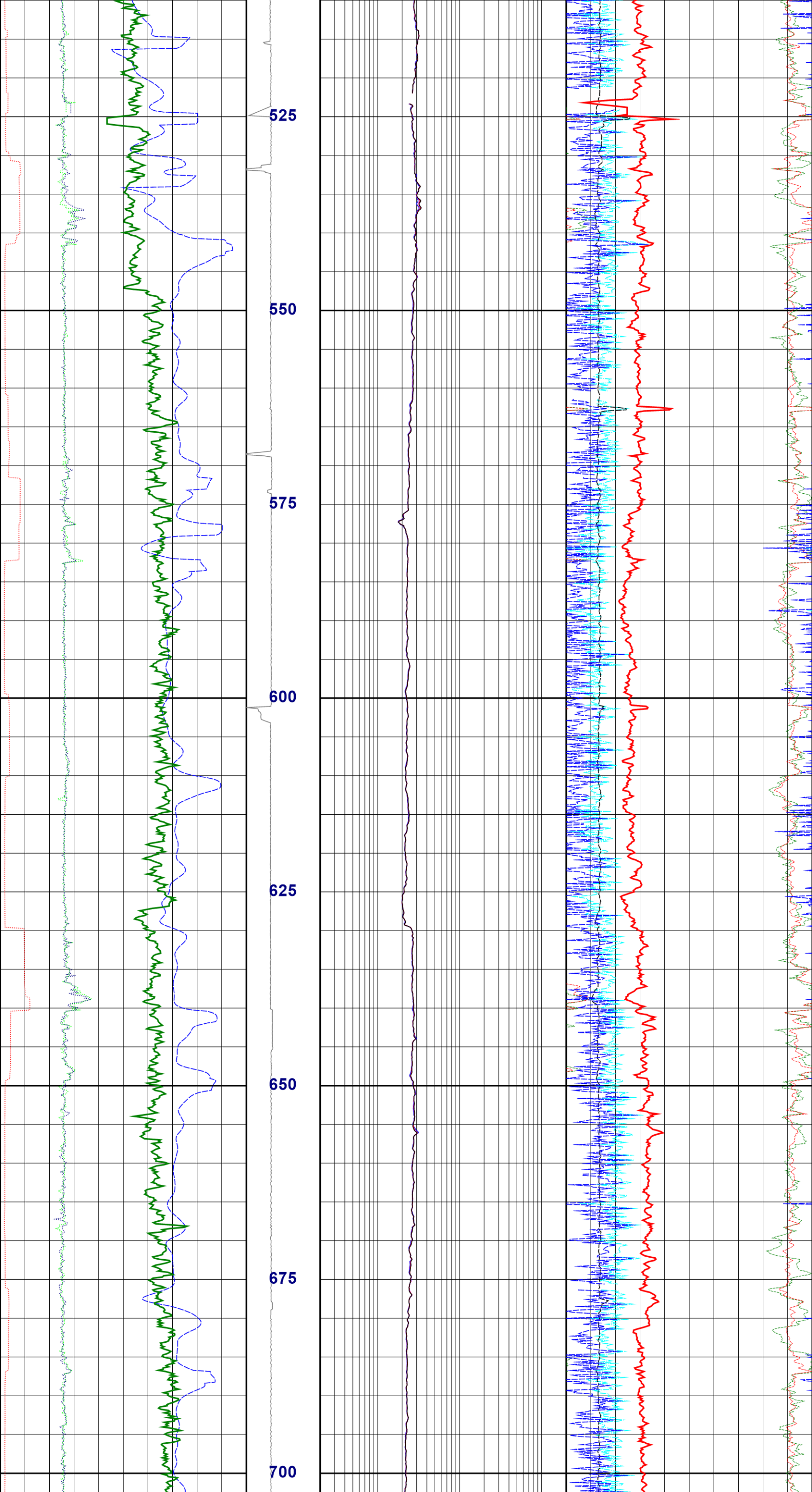
MWD spliced data including:
MWD ECOSCOPE, run 2 (8.5 in pilot)
MWD ARC, run 3 (26 in)
MWD ARC, run 5 (12.25 in pilot)
MWD ARC, run 8 (12.25 in)
MWD ARC-RAB, run 9-13 (8.5 in)

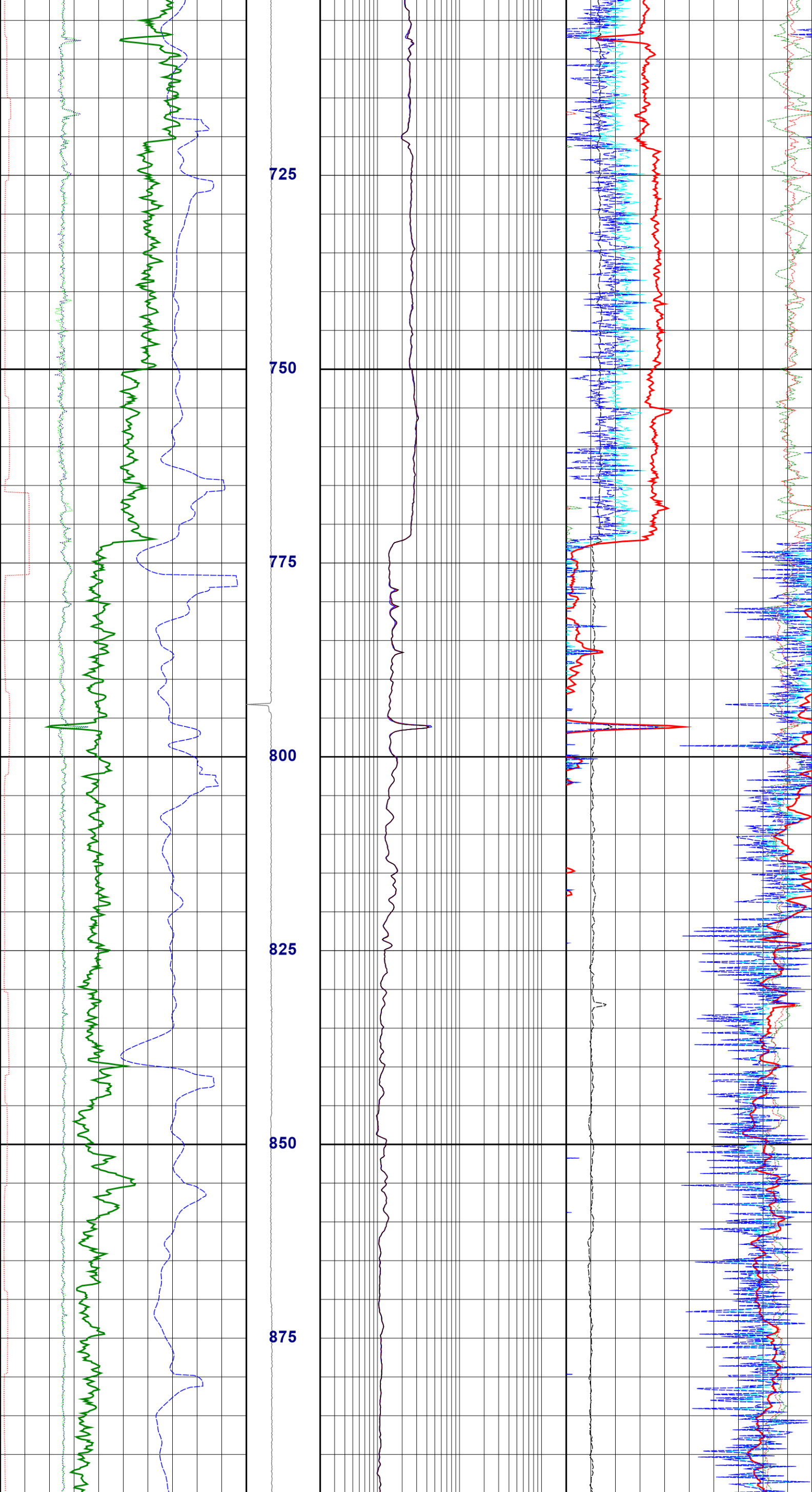
Splice depths:
Run 3-2:
GR spliced at 392.7 m.
ARTM and ROP5 spliced at 400.7 m.
Run 2-5:
GR spliced at 934.2 m.
ARTM and Resistivities spliced at 933.3 m.
ROP5 spliced at 944.1 m.
Run 5-8:
GR, ARTM and Resistivities spliced at 1896.9 m.
ROP5 spliced at 1906.2 m.
Run 8-(9-13):
GR spliced at 2574.3 m.
ARTM spliced at 2575.0 m.
ROP5 spliced at 2583.9 m.

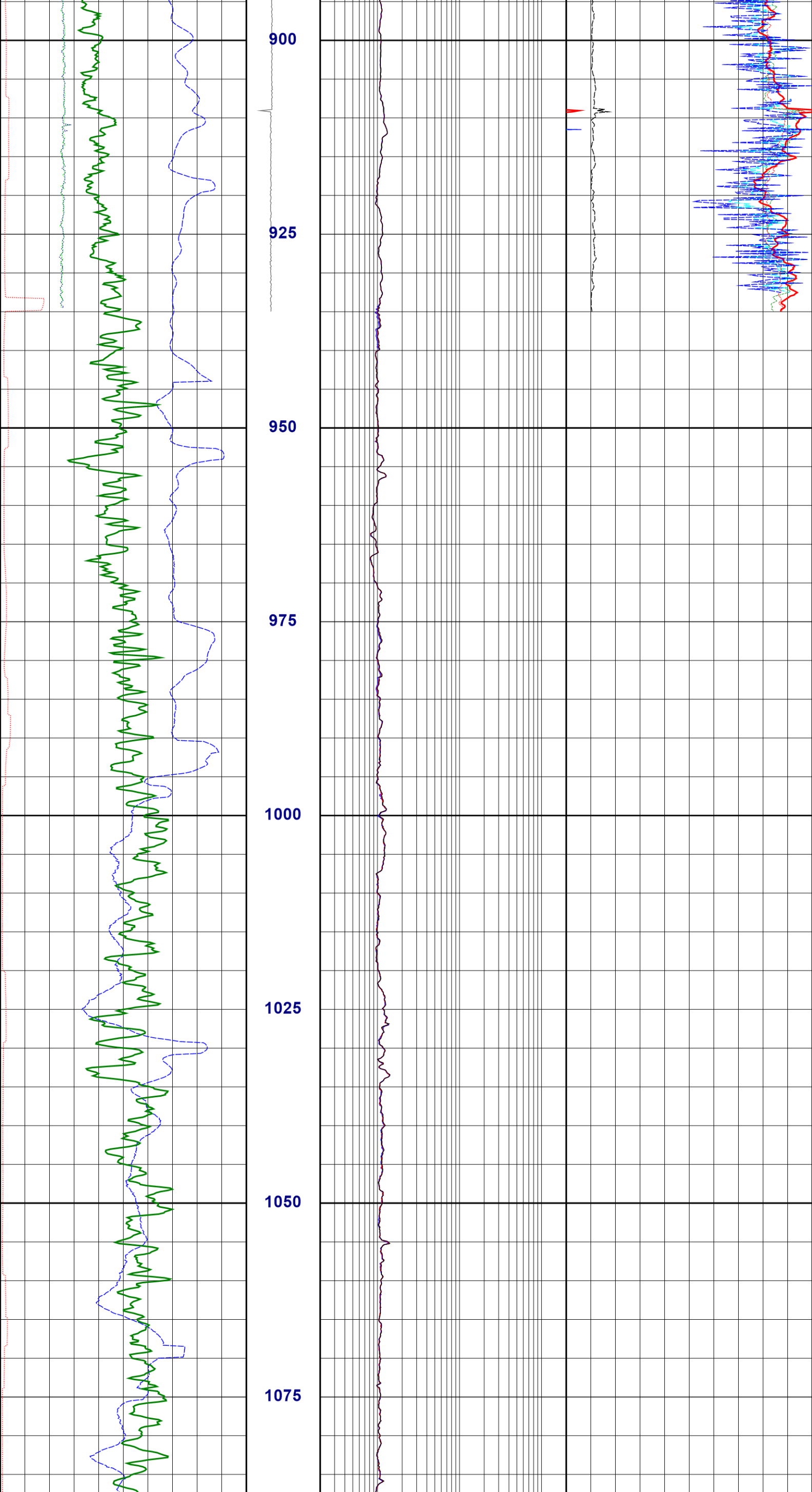
Editing:
Run 9: Resistivities removed in casing above 2583.0 m.

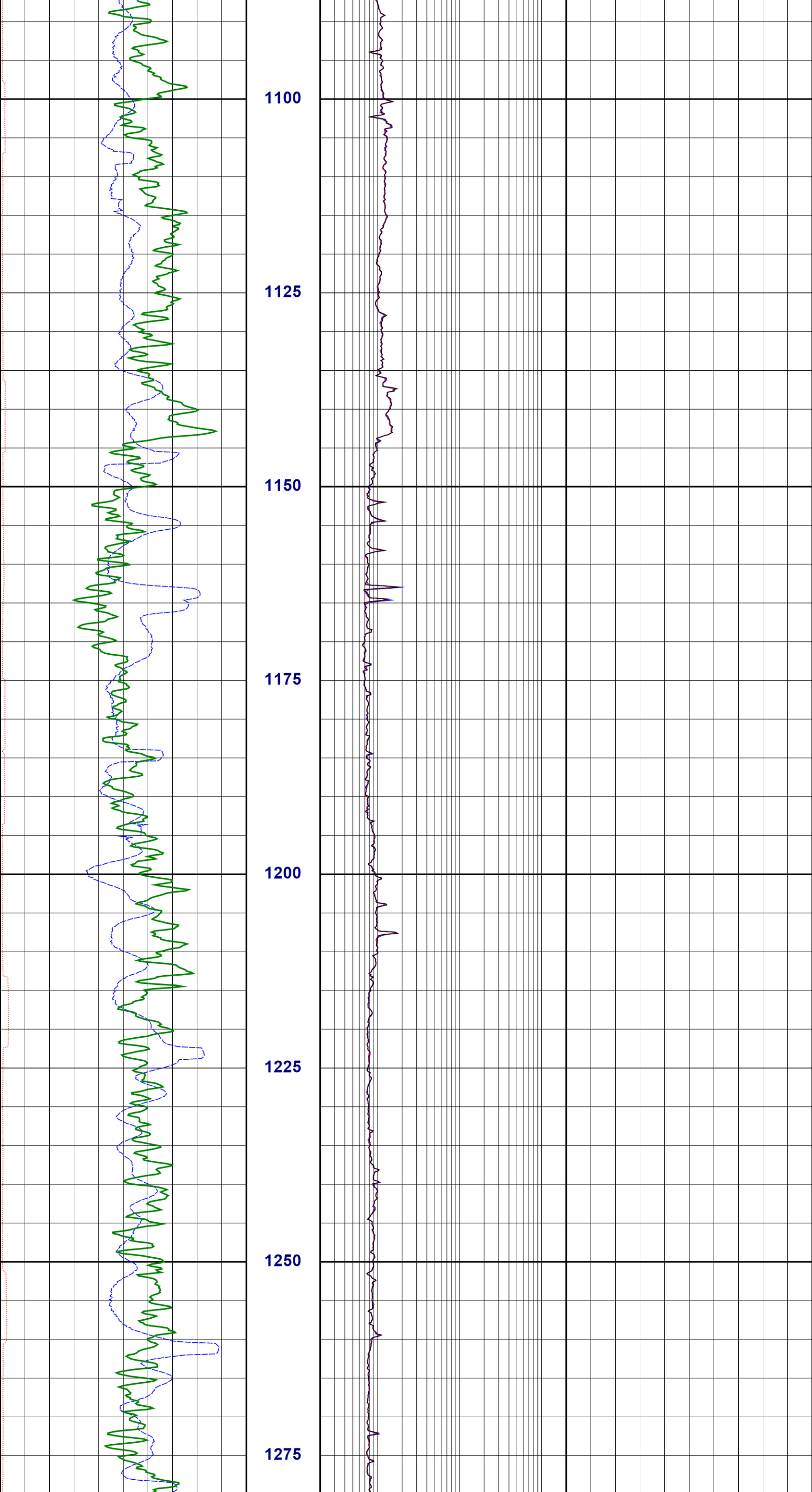
Log Description	
ARTM	ARC Resistivity Time After Bit (TAB_ARC_RES+ARTM)
UCVE	Ultrasonic Caliper, Vertical Diameter
UCHO	Ultrasonic Caliper, Horizontal Diameter
ROP5	Rate of penetration averaged over the last 5 ft (1.5 m)
GR	Gamma Ray (GRMA+GR_ARC_FILT+GR_ARC)
P16H	Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected.
P22H	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected.
P28H	Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected.
P34H	Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected.
P40H	Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected.
DRHU	Bulk Density Correction, Up
BPHI	Best Thermal Neutron Porosity, Average
DRHO	Bulk Density Correction

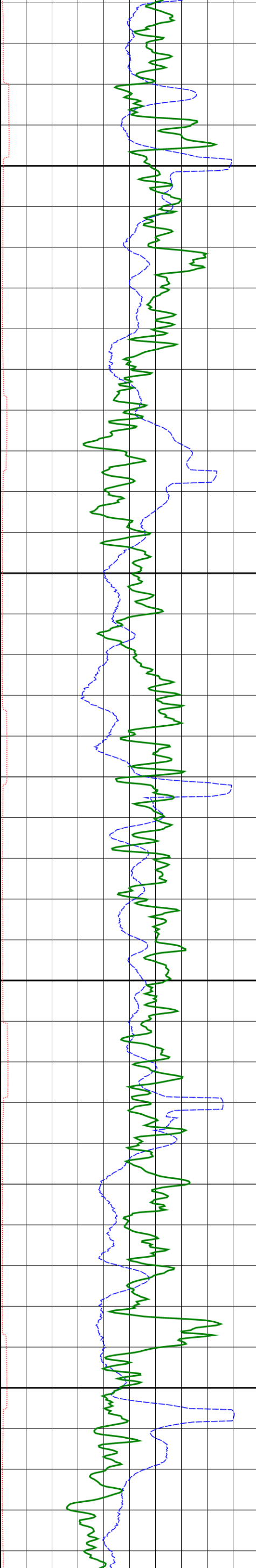
0.0	ARTM (h)	10.0	0.0 CRPM (c/min)	300.0	0.2	P16H (ohm.m)	200.0	-0.8	DRHU (g/cm3)	0.2
6.0	UCVE (in)	16.0	DEPTH		0.2	P22H (ohm.m)	200.0	45.0	BPHI (pu)	-15.0
6.0	UCVO (in)	16.0	M		0.2	P28H (ohm.m)	200.0	-0.8	DRHO (g/cm3)	0.2
200.0	ROP5 (m/h)	0.0	1:500		0.2	P34H (ohm.m)	200.0	0.0	PEF	20.0
0.0	GR (gAPI)	150.0			0.2	P40H (ohm.m)	200.0	1.95	RHOB (g/cm3)	2.95
								0.95	RHOB (g/cm3)	1.95
								45.0	TNPH (pu)	-15.0











1300

1325

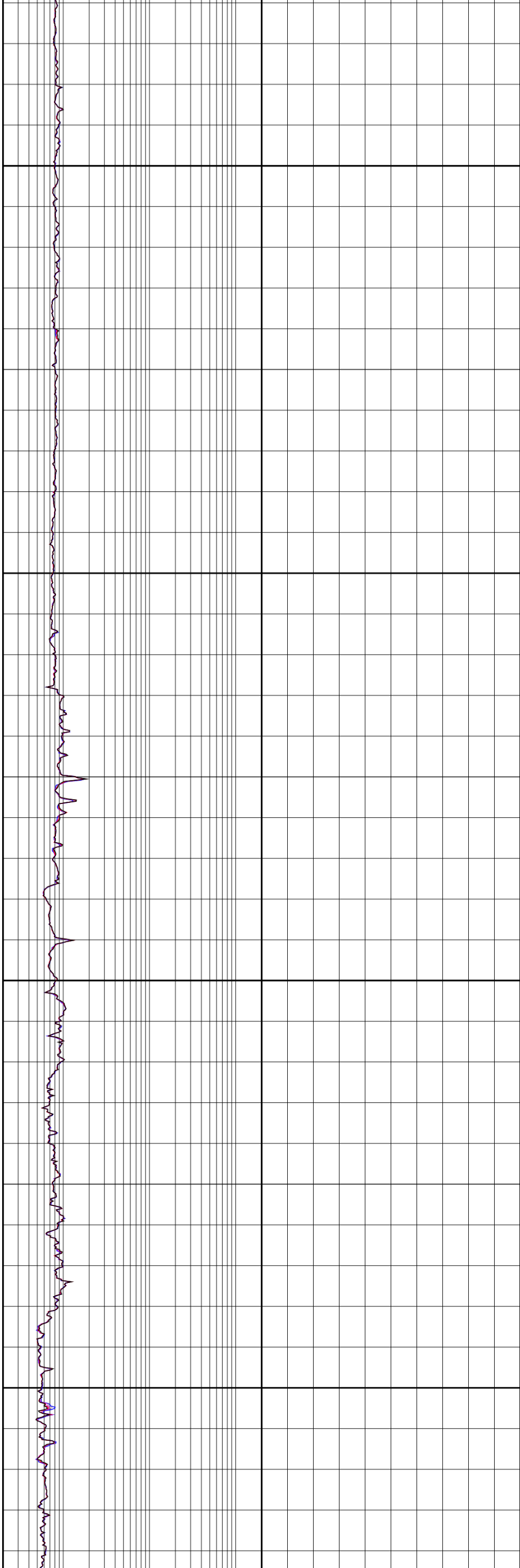
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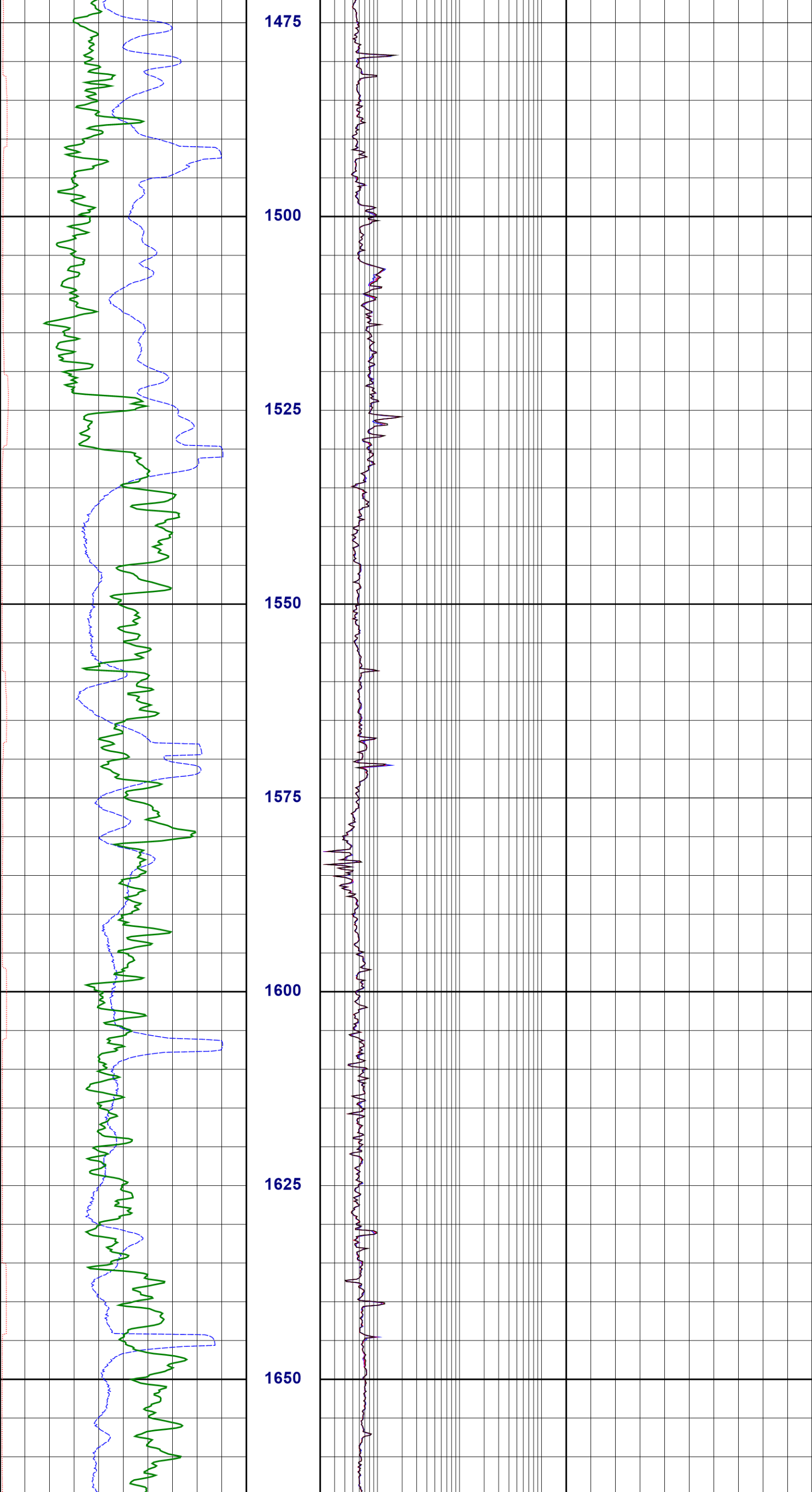
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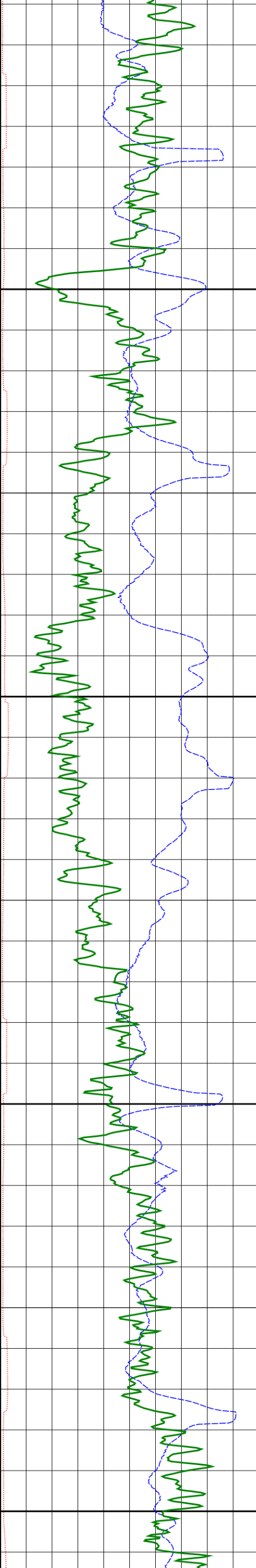
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1425

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1675

1700

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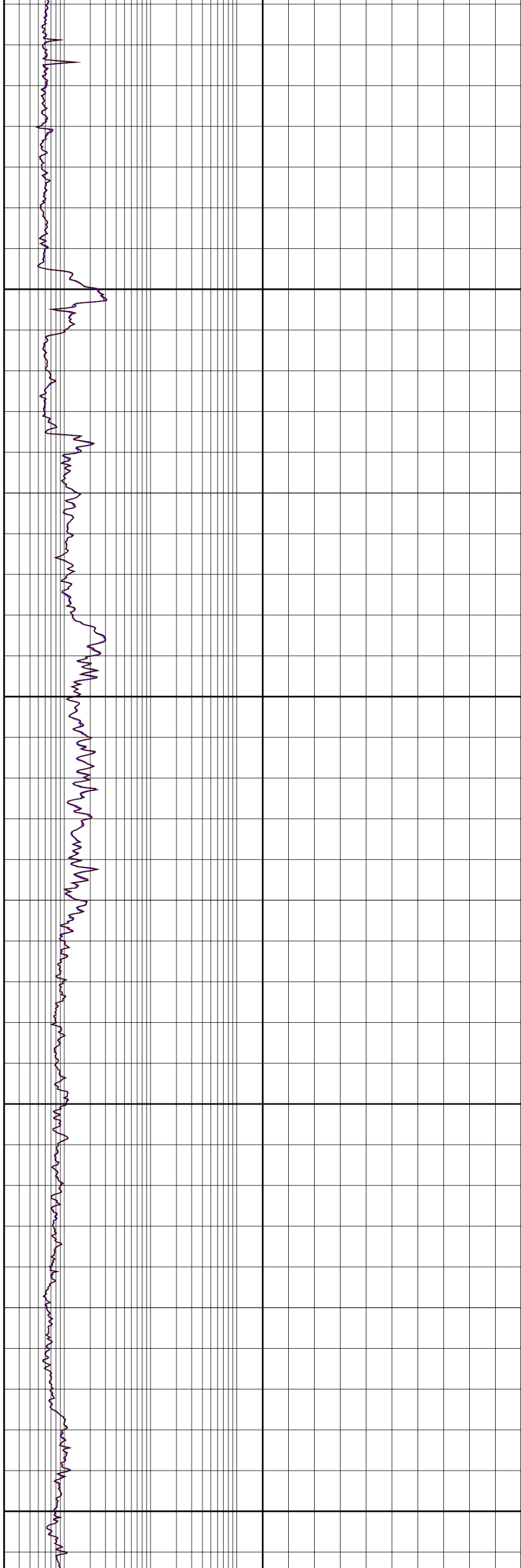
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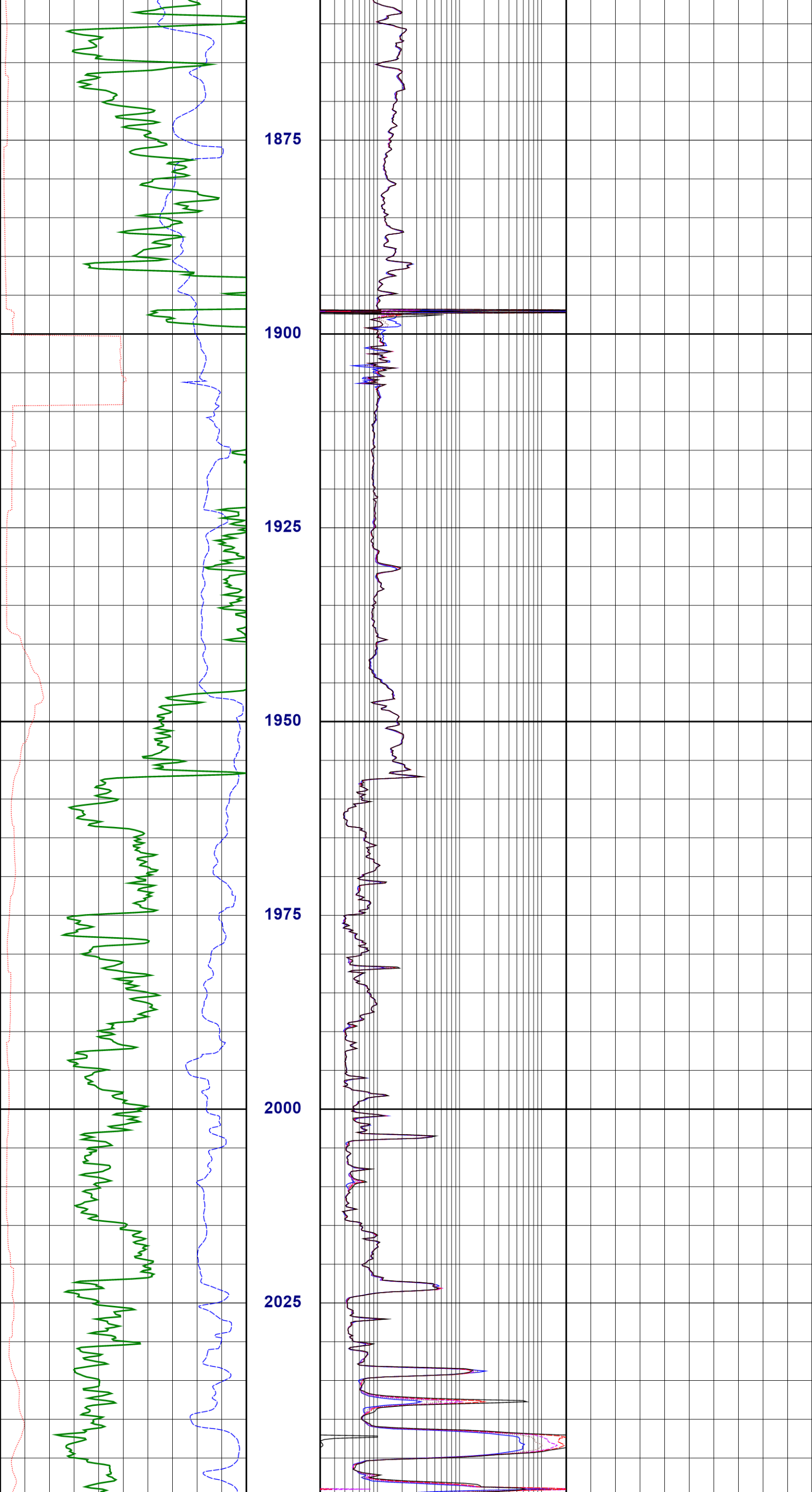
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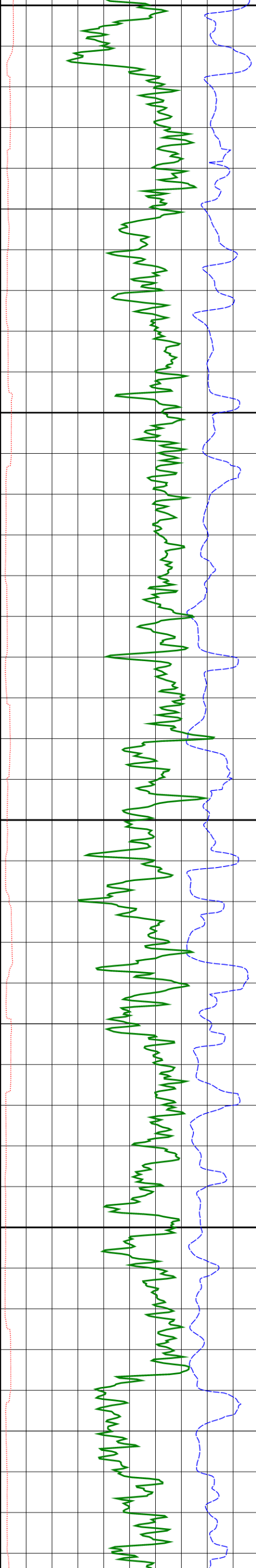
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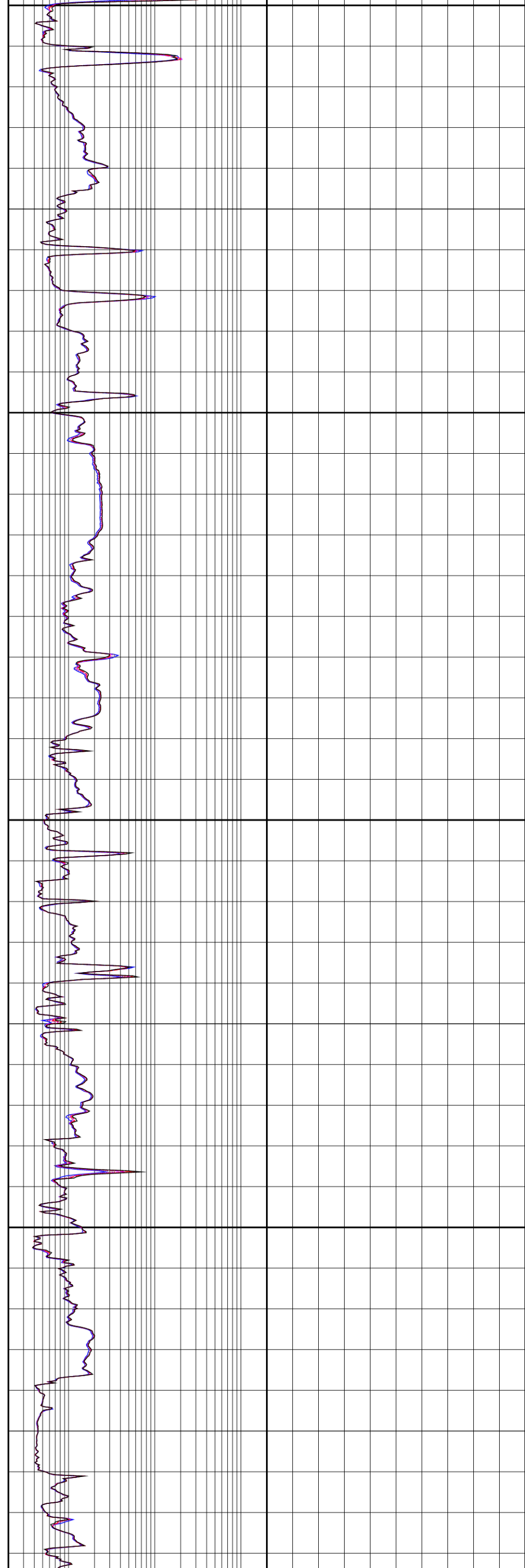
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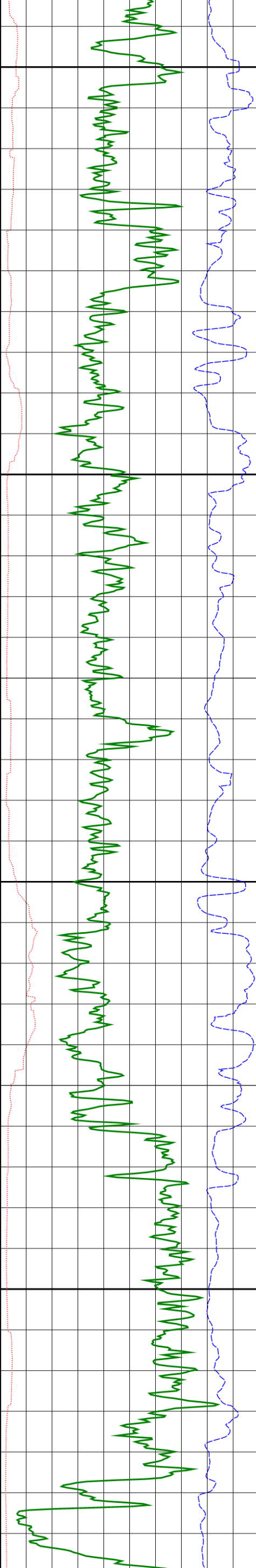






2050
2075
2100
2125
2150
2175
2200
2225





2250

2275

2300

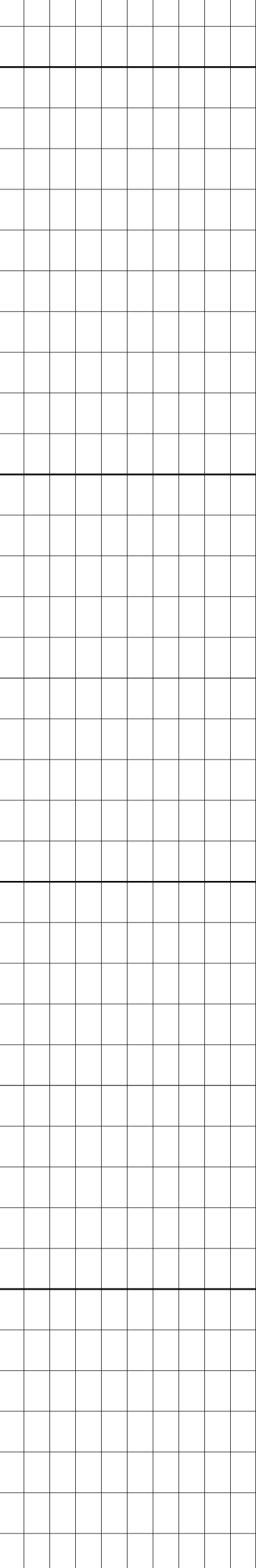
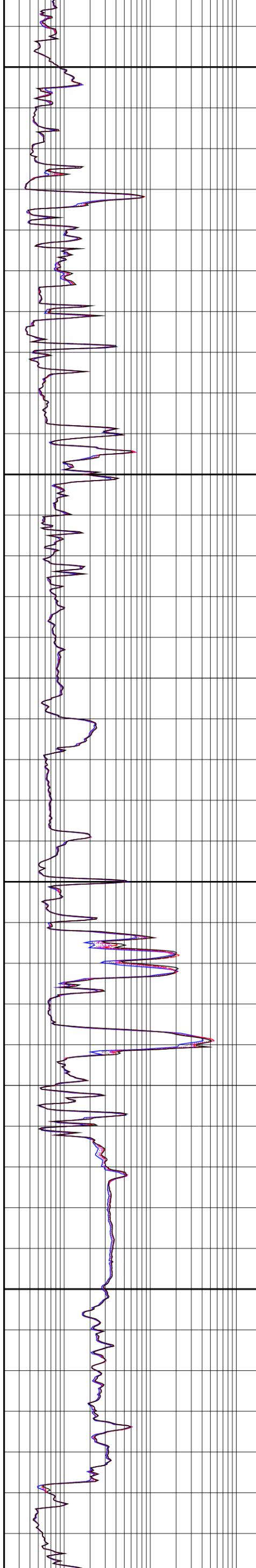
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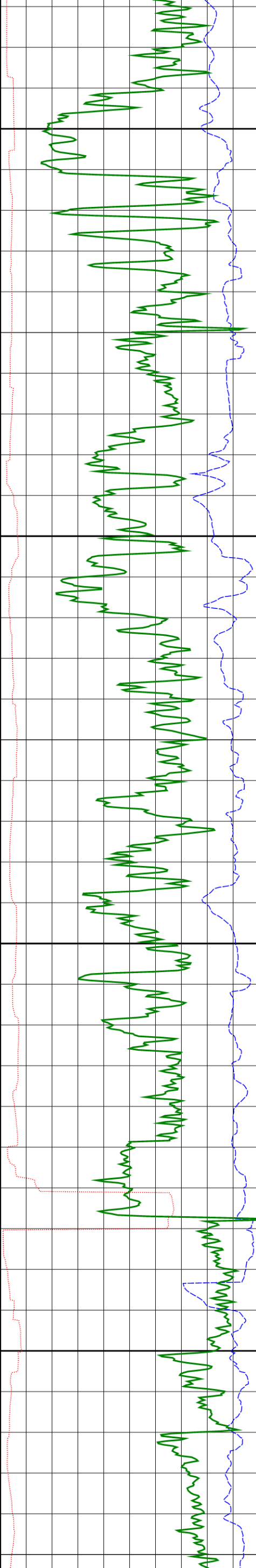
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2375

2400

2425





2450

2475

2500

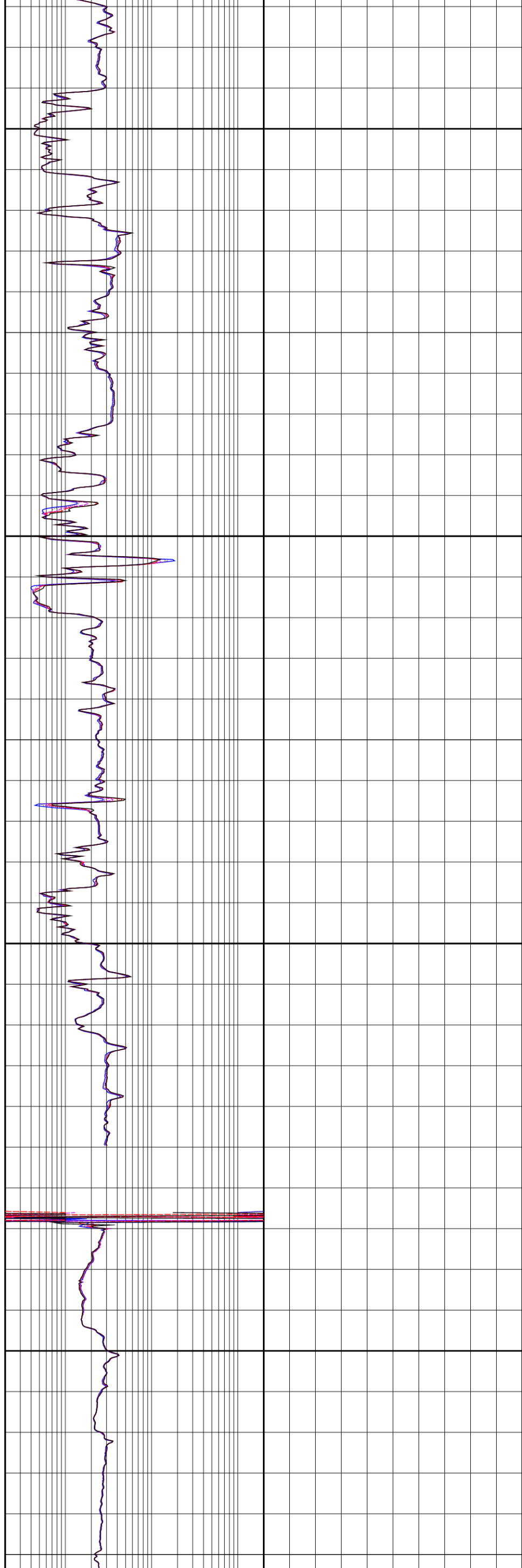
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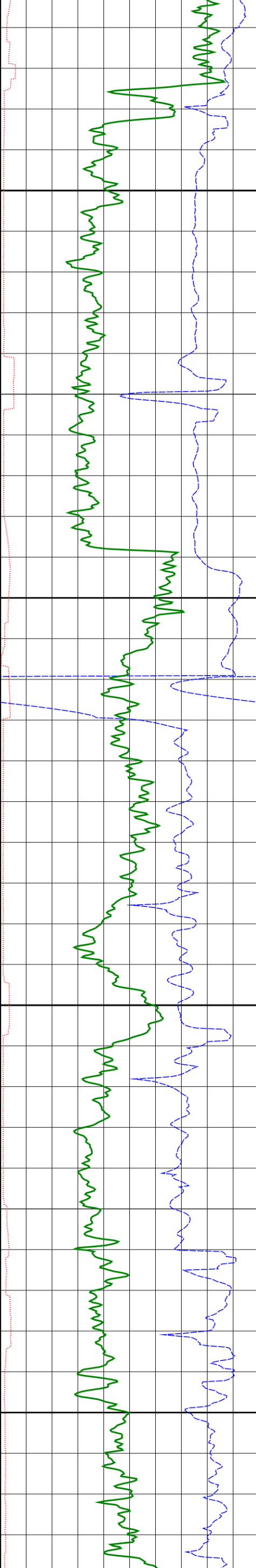
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2600

2625





2650

2675

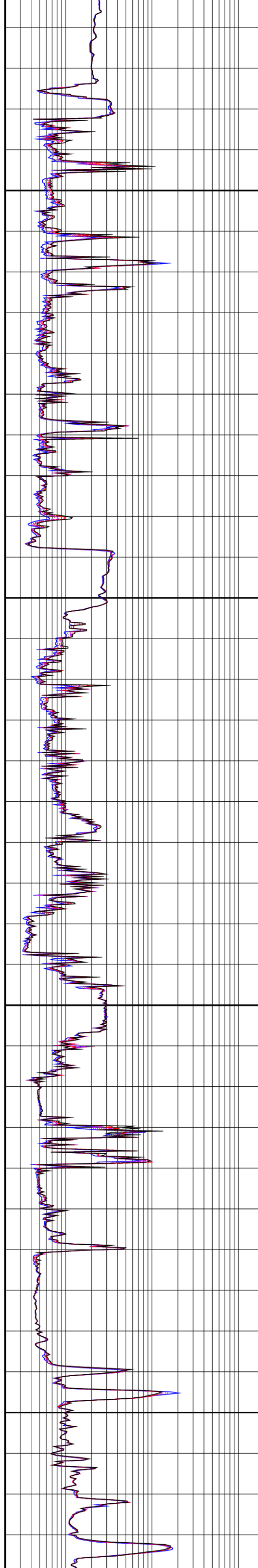
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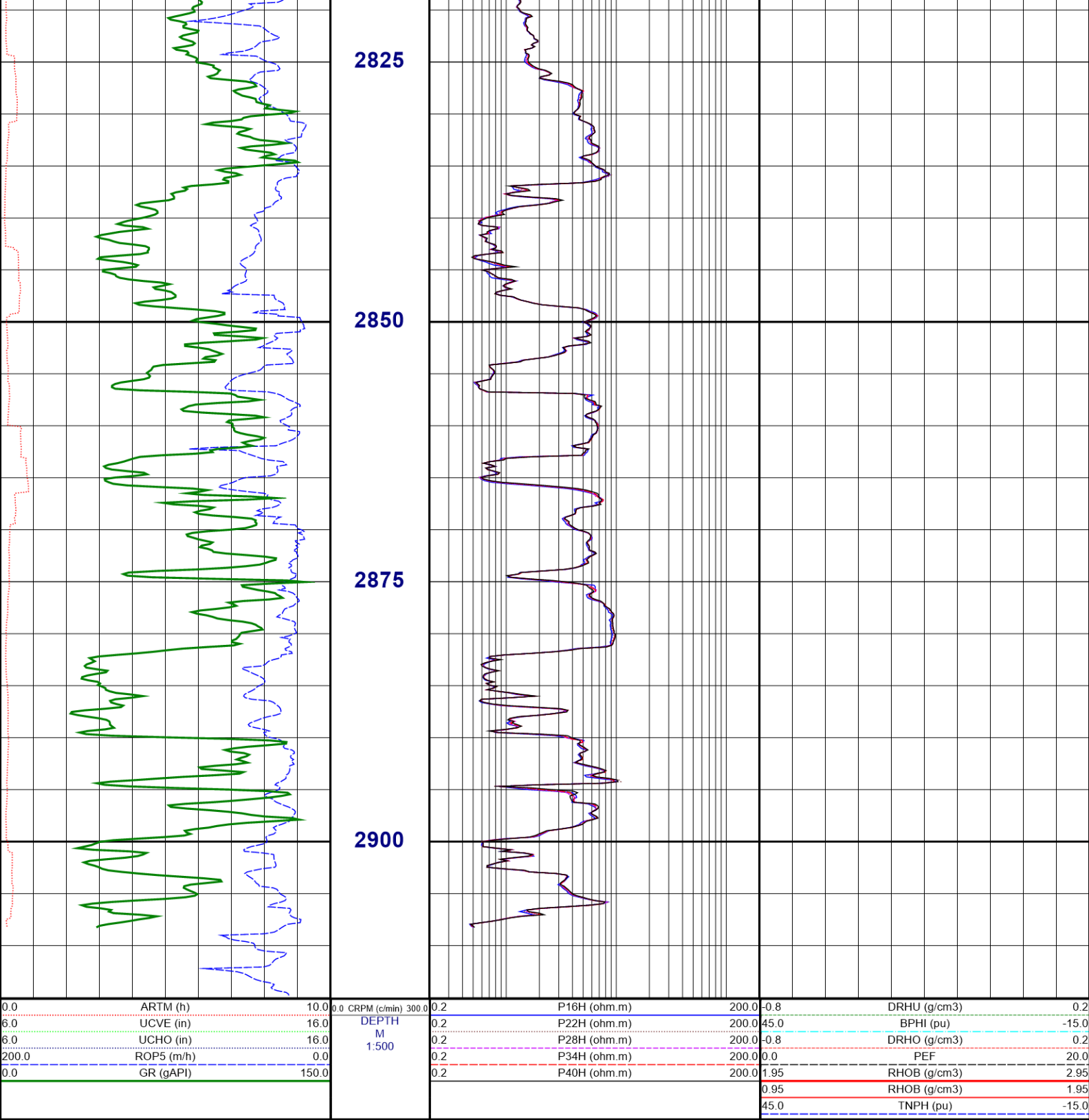
2725

2750

2775

2800





Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

